

How to control the crystallization of metallic glasses during laser powder bed fusion? Towards part-specific 3D printing of *in situ* composites

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ARTICLE INFO

Keywords:

Bulk metallic glasses (BMGs)
Laser powder bed fusion (LPBF)
Selective laser melting (SLM), *in situ* composites
Crystallization, 4D-printing

ABSTRACT

This paper describes a strategy for creating highly oriented crystalline-amorphous composites using the laser powder bed fusion (LPBF) process. The strategy involves using a novel two-stage melting approach and ultra-high-pressure hot isostatic pressing (HIP) on well-known AMZ4 ($Zr_{59.3}Cu_{28.8}Al_{10.4}Nb_{1.5}$) and equiatomic CuZr amorphous alloys.

The experiments demonstrate that by the fine-tuning laser parameters, allowed to obtain parts with purely amorphous material and to create geometry-specific microstructural design composites based on laminate amorphous-crystalline structure. This approach also provides novel opportunities for nonequilibrium phase distribution design by controlling local crystallization in the heat-affected zone (HAZ) and avoiding heat accumulation. Additionally, the porous amorphous material can be densified without crystallization using HIP at a temperature near the supercooled liquid region.

The distribution of the crystalline phase created during LPBF and crystallization on pre-induced nuclei during HIP was proven to be a critical factor for composite properties. Wear and bending tests reveal the influence of crystalline-amorphous layers orientation on mechanical properties. The functional demonstrators were manufactured to show the possibilities in the design for additive manufacturing (DfAM) with a microstructure-designed composites.

1. Introduction

Since the discovery of metallic glasses (MGs) in the 1960 s, mostly their soft magnetic properties have been broadly implemented in the industrial environment [1]. Despite the extraordinary mechanical and

chemical properties of MGs, their use as structural materials is still limited by their poor ductility and complex manufacturing processes [2], with only a few companies, such as Amorphology [3] and Heraeus Amloy [4], offering MG based products. Injection molding of MGs requires expensive tooling and high-purity feedstocks. Although

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<https://doi.org/10.1016/j.addma.2023.103775>

Received 25 May 2023; Received in revised form 3 August 2023; Accepted 6 September 2023

Available online 7 September 2023

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thousands of interesting compositions were invented and announced, only a few have the functional glass-forming ability (GFA), toughness, and processability high enough to be formed into a product of viable size and at a reasonable price [5,6]. Similar constraints applies to metallic glass matrix composites (MGMCs), which, despite their enhanced mechanical strength and tensile ductility achieved through *in situ* and *ex situ* methods [7], have attracted significant scientific attention, although no widespread MGMC products were implemented.

Additive manufacturing (AM) techniques, such as laser powder bed fusion (LPBF), have the potential to revolutionize the way products are designed and manufactured. To fully realize the potential of AM, it is essential to take a holistic approach to design parts and products specifically for the constraints and capabilities of these processes. This approach, known as Design for Additive Manufacturing (DfAM), is crucial for the optimization of mechanical properties and functional performance of the final product. In this paper, we focus on the application of DfAM in the field of bulk metallic glasses (BMGs) for LPBF. BMGs have excellent mechanical and chemical properties, but their use as structural materials is limited by their poor ductility and demanding manufacturing processes. DfAM can help to overcome these limitations and enable new possibilities for the use of BMGs in structural applications.

The accelerated development of Additive Manufacturing (AM) techniques, especially LPBF [8,9] has attracted considerable attention in material science. LPBF, with its rapid heating and cooling rates [10], has been proven to generate metastable microstructures and has been employed in numerous MGs, including zirconium, titanium, and iron-based systems [11–13]. The traditional manufacturing limitations of MGs, such as poor machinability and maximum casting diameter, can be overcome via LPBF, leading to new commercial applications. However, laser processing of MGs poses challenges.

Due to the metastable nature of MGs, LPBF optimization compromises both porosity and crystalline phase content control [14–16]. In a conventional single-exposure LPBF melting strategy, MGs are subjected to a disadvantageous trade-off where the amorphous phase content opposes the material density. The abovementioned requirements are contradictory as the higher energy density (ED) used in the LPBF process decreases the porosity but accelerates devitrification in the heat-affected zone (HAZ), which is usually considered a material defect. The material processing window becomes very narrow as high ED results in severe devitrification, while lower ED induces lack of fusion porosity [17–19].

This trade-off is well documented in the MG processing literature [11,12] and has been addressed in our previous paper [17] and a commercialized patent [18]. Exposure features two different melting modes - exposure in conduction and keyhole mode for material densification and amorphization, respectively [19]. Separation of scanning lines allow to maintain high cooling rates obtained due to keyhole mode melting by reducing the local heat accumulation resulting in the increasing the amorphous content. This innovative scanning strategy provides microstructural design freedom, independent of powder densification issues, as the thermal accumulation influence of adjacent melting tracks has been minimized. On the contrary, most other studies on parameters rely on basic scanning strategies and changes in parameters such as laser power, scanning speed, and hatch distance [11,12,20].

Despite the negative influence of devitrification on the mechanical properties of BMGs, several examples of MGMC with improved mechanical properties were obtained with both *ex situ* [21,22] and *in situ* [23–27] LPBF. Nevertheless, due to the uncontrolled devitrification of MGMCs in the HAZ, the amorphous-crystalline structure results from the manufacturing parameters and cannot be independently controlled, [27] resulting in either lack-of-fusion defects or severely crystallized materials [28]. Although LPBF offers high heating and cooling rates, uncontrolled crystallization remains a major concern. Ouyang et al. [29] demonstrated that crystallization primarily occurs due to repetitive heating in the HAZ, and the printability of certain alloys is determined

by their resistance to devitrification under flash heat treatment conditions.

This research aims to develop materials with a precisely designed distribution of amorphous and crystalline phases. We introduce a strategy for creating fully dense spatially controlled MGMCs and investigate the impact of phase distribution design on the properties of the obtained products. Control of both local crystallization in the heat-affected zone and global overheating has been achieved. This approach allows for part-specific microstructural control and the generation of novel design opportunities for LPBF technologies.

The paper is structured in a way that enables other teams to replicate the results with any glass-forming alloy, starting with a single-track experiment, optimizing the melting strategy, conducting heat treatment, and performing high-throughput characterization of the final product.

AMZ4 material was chosen as the main material for the study due to its commercial availability on the market in form of powder, which allowed for a sufficient amount of testing. Subsequently, tests were performed on the designed CuZr alloy to verify the developed printing technique.

In conclusion, this study underscores the potential of DfAM in creating highly oriented crystalline-amorphous composites with improved mechanical properties and novel opportunities for micro-structural design using the laser powder bed fusion process.

The paper consists of a practical guide on providing spatially controlled MGMCs answering the question: how to control the crystallization of MGs and MGMCs processed via LPBF?

2. Experimental

2.1. Materials preparation

The experiment initially utilized the well-documented amorphous alloy AMZ4 (at. $Zr_{59.3}Cu_{28.8}Al_{10.4}Nb_{1.5}$) produced by Heraeus GmbH [15,30,31]. Due to sufficient GFA, high toughness, and low cost, the alloy is considered a promising feedstock for powder metallurgy applications such as LPBF.

The materials for the experiment were selectively laser melted with Realizer SLM 50 machine, equipped with a maximum laser power of 120 W and a 20 μm diameter focus. The oxygen concentration within the chamber was maintained below 0.1%. Despite the relatively low laser power, an exceptionally high power density is achieved due to the small focal spot diameter. This is beneficial for the process as a high power density is correlated with keyholing and high heating rates [17].

LPBF process optimization was divided into three distinct phases: 1. single tracks, 2. scanning strategy validation, and 3. laser parameters. To understand the fundamental behavior of the material under laser melting conditions, the first phase was performed on bulk material. The AMZ4 powder was arc-melted and cast into a disc with a diameter of 25 mm and thickness of 2 mm. After grinding, the cast material underwent laser remelting in a single-track format, with alterations made to laser power and laser speed parameters. The second and third phases of the LPBF experiments involved AMZ4 powder melting on a Ti-6Al-4V build plate with a constant layer thickness of 25 μm . The second phase, investigated the influence of hatch separation and delay between adjacent tracks on the microstructure. During the third phase, the influence of various laser speed parameters and hatch distance was studied. Given that the outcomes of previous phases determine the parameters of the process in subsequent phases, the details have been listed in the Results section.

The final densification step was accomplished using HIP. Process was conducted using following parameters: heating rate 40 K/min, holding temperature of 420 $^{\circ}C$, holding time of 360 s, and pressure of 5.0 Ar 1 GPa. Parameter selection was based on the previous preliminary studies to simulate the SPS conditions in which it was possible to obtain amorphous AMZ4 gears [32]. The gas pressure was set to the maximum

value of the used equipment.

After the LPBF optimization phases, standardized parameters (AMORPHOUS, AMORPHOUS+HIP, COMPOSITE, COMPOSITE+HIP) were chosen for comprehensive mechanical characterization to limit specimen quantity and examine different microstructures.

2.2. LPBF simulation

A finite element (FE) model based on the commercial Abaqus package was applied to evaluate different laser scanning strategies and predict the thermal history during the laser melting process. In order to effectively simulate the thermal history under different process conditions, the melt pool dynamics including mass loss, recoil pressure due to evaporation, and fluid flow in the melt pool were not simulated. It is clear that fluid flow is critical to physically predict the melt pool evolution with accuracy, in particular, for keyholing mode. However, it is computationally expensive. To consider its impact to the heat transfer in the melt pool, the heat transfer coefficient is artificially increased to enhance the heat conduction [33]. Note that in this study the FE model is not intended to accurately predict the thermal history, but to understand the thermal behavior under different scanning strategies to assist in explaining the experimental observations. Therefore, the current model adopts the Goldak heat source with only heat transfer model without the consideration of the powder layer, which provides reasonable accuracy for the evaluation of thermal history without the consideration of fluid flow in laser welding [34,35] and LPBF [36].

The heat diffusion equation was solved for the transient thermal field:

$$\rho C_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q$$

Where ρ is the mass density and specific heat capacity, T is the temperature, k is the thermal conductivity. Q is the heat source that includes the laser energy and the latent heat of fusion.

The experimental data for the laser power of 120 W and speed of 2000 mm/s from the single-track experiments were used to determine the coefficients. A half symmetry domain with the size of $0.6 \times 0.1 \times 0.25 \text{ mm}^3$ was used for the single-track simulation. The symmetry plane was along the center of the laser scanning direction. A fine mesh size of $5 \mu\text{m}$ was used to ensure accuracy [36,37]. Temperature-dependent thermophysical properties (thermal conductivity, specific heat, density, and latent heat of phase change) for AMZ4 were used in the simulations, as shown in previous reports [38–40]. A heat transfer coefficient of $10 \text{ W/m}^2\text{K}$ was applied to all side surfaces and top surfaces. An emissivity of 0.35 was used on the top surface to account for radiation. With the confirmed double-ellipsoidal heat source model, two laser scanning strategies were simulated with the same power and speed from the calibrated model: 1) no delay between laser scanning, and 2) 1 s delay. The selection of these two cases was to understand the differences in thermal history with the effect of laser delay time. A domain size of $1 \times 1 \times 0.25 \text{ mm}^3$ was applied in this multiple-line scanning simulation with a hatch spacing of 0.1 mm. The same boundary conditions and mesh size were employed as in the single-track simulation. In this setup, 50 scanning lines were simulated to allow the heat transfer to achieve a quasi-steady state.

2.3. Characterization techniques

2.3.1. Light microscopy (LM)

To reveal crystallites in LM samples were etched with Kroll etchant. A Nikon Eclipse MA200 microscope equipped with Nomarski contrast was used to capture the images. The final composite gear and knife image was created via image stitching with Keyence VHX 5000 microscope. In the differential chemical resistance between crystalline and amorphous phases rendered crystalline regions dark post-etching due to

their marked susceptibility to the etchant.

2.3.2. X-ray diffraction (XRD) analysis

The third phase of LPBF parameters optimization utilized a Rigaku Miniflex II with $\text{CuK}\alpha$ source within a $30\text{--}45^\circ$ 2-theta range to validate the existence of crystalline phases. Final XRD measurements on standardized parameters (AMORPHOUS, AMORPHOUS+HIP, COMPOSITE, COMPOSITE+HIP) were conducted using the HZG beamline (P07B) located at PETRA III (a 6 GeV energy electro-storage ring with a 100 mA beam current) at DESY, Hamburg, Germany. The high energy of synchrotron radiation ($\lambda = 0.142342 \text{ \AA}$) allowed phase analysis in transmission geometry. The bulk specimens of dimensions $10 \times 10 \times 1 \text{ mm}^3$ were scanned using the continuous mode [41], and the area Mar345 Image Plate detector with a beam size of $0.9 \times 0.9 \text{ mm}^2$. To avoid the effect of crystallographic texture a rotation by 180° was applied [42,43]. Subsequently, the obtained data were integrated by Fit2D software and shown in a graph as a function of intensity versus 2Theta angle. For calibration, a LaB_6 powder sample was used. The volume fraction of the crystalline phase was assessed using the HighScore Plus software.

2.3.3. Electron microscopy (EM)

Preliminary AMZ4 powder characterization was carried out using Hitachi SU 3500 microscope at 20 kV while bend samples fractography (in a secondary electron mode, SE) and microstructural observations (in a backscattered electron mode, BSE) were carried out using Hitachi SU 8000 microscope at 5 kV. Samples for microstructural observations were ion milled using Hitachi Ion Milling System IM4000.

TEM lamellae were fabricated using a focused ion beam (FIB) Hitachi NB5000. The lamellae were extracted from a benchmark COMPOSITE specimen in the HAZ and then observed using a JEOL JEM 1200 with a Gatan Orius camera at an accelerating voltage of 120 kV.

2.3.4. Differential scanning calorimetry (DSC)

Differential scanning calorimetry was performed using a DSC Perkin Elmer DSC 8000 at a standard heating rate of 40 K/min (ranging from 293 K to 973 K) within an Ar 5.0 flow protective atmosphere and an alumina crucible. This technique was used to quantify the content of the glassy phase. Samples weighing $20 \pm 2 \text{ mg}$ were obtained from the center of a standard $10 \times 10 \times 3 \text{ mm}$ sample to avoid contamination from the build plate.

2.3.5. Computed tomography (CT)

Computed tomography examinations were performed on an Xradia XCT-400 at 80 kV voltage, resulting in a voxel size of $4 \mu\text{m}$. Benchmark parameters were selected for this investigation with the specimen size of $10 \times 10 \times 1 \text{ mm}$.

2.3.6. Mechanical tests

The procedure for the 4-point bending tests on benchmark parameter specimens was as reported previously by Bordeenithikasem et al. [15] for LPBFed AMZ4. The force was applied perpendicular to build direction on $2 \times 3 \times 25 \text{ mm}$ specimens and either perpendicular (90°) or parallel (0°) to the direction of composite layers.

The wear resistance influence of composite structure orientation was evaluated on 25 mm diameter COMPOSITE disk-shaped specimens. These were tested for abrasion with a $1/4"$ diameter zirconium oxide ball. The tests were performed on a Tribotester T-21 ball-disk dry tester with the friction coefficient recording.

A nanoindentation tester (Anton Paar NHT3) was used to map the hardness and elastic modulus of the COMPOSITE specimen. The series of 961 nanoindentation tests (31×31 grid with $5 \mu\text{m}$ spacing) were carried out at room temperature with an applied load of 4 mN, 8 mN/min loading and unloading rates, and 10 s pause. The analysis was performed using the Oliver-Pharr procedure.

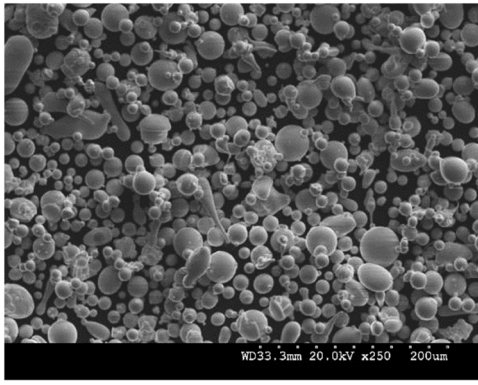


Fig. 1. SEM images of the AMZ4 powder.

Table 1
Parameters of single-track experiments and dimensions of melt pools.

Laser speed [mm/s]	Power [W]	Energy density [J/mm ³]	Melt depth [μm]	Melt width [μm]
3500	120	0034	118	71
3000	120	0,04	143	85
2500	120	0048	167	93
2000	120	0,06	198	96
1500	120	0,08	250	117
300	30	0,1	103	146
250	30	0,12	113	163
200	30	0,15	116	176

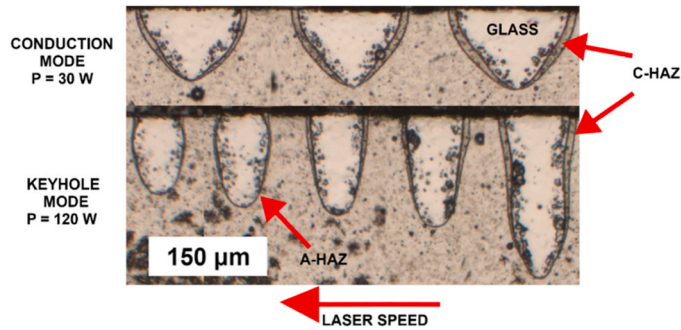


Fig. 2. Microstructure of partially crystalline cast AMZ4 re-melted with various laser parameters. A-HAZ – Amorphous heat affected zone, C-HAZ crystalline heat-affected zone.

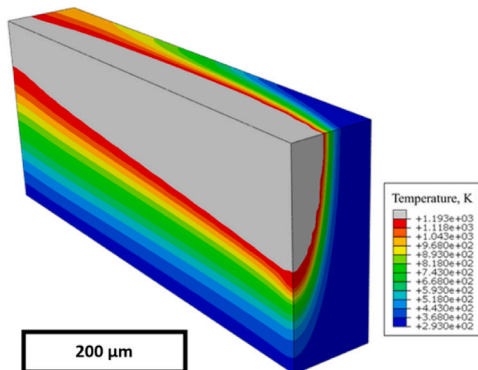


Fig. 3. Simulation of the temperature distribution in the melt pool at 120 W/2000 mm/s.

2.3.7. Oxygen and nitrogen analysis

The oxygen concentration was measured by LECO tester using the hot extraction NO analyzer on virgin powder, pre-melted material, and benchmark parameters specimens (AMORPHOUS, COMPOSITE). The provided values represent the mean of 3 measurements.

3. Results and discussion

3.1. Powder characterization

The morphology of the virgin spherical powder s shown in Fig. 1. The particle size distribution, DSC and XRD results of the AMZ4 powder are reported in the previous article from our research group [32]. As the powder was produced via electrode inert gas atomization (EIGA), several deformed and non-spherical particles are present. Due to the small radius of the rod-like particles the sieving method would not separate them from the powder with high sphericity. The concentrations of oxygen and nitrogen in the powder, as measured by the LECO ON analyzer, were found to be 1762 ± 40 ppm wt. and 47 ± 5 ppm wt. respectively. Given its high oxygen content and industrial-grade elements, the alloy is considered a cost-effective BMG alloy, though it is relatively poor glass former [44,45]. However, the XRD analysis confirms the amorphous structure of the materials in the as-received state, as reported previously using the same batch of powder [32].

3.2. Single-track experiments – local crystallization

The first experiment was performed on bulk material to understand the fundamental behavior of the AMZ4 material under laser melting conditions. Following the surface grinding, the ingot was laser remelted with single-track varying laser power and speed. Summary of the laser parameters used and dimensions of the resulting melt pools are given in Table 1, while the microstructure of the remelted material is depicted in Fig. 2. Due to the low purity of commercial-grade AMZ4 and high oxygen concentration in the powder, GFA was limited, and the ingot was partially crystallized. The white featureless regions are amorphous melt pools (MP) obtained by vitrification in a single laser pass, while dark etched regions are the crystalline heat-affected zones (HAZ). High laser power (120 W) provides melt pools with a high depth-to-width ratio, which is indicative of partial material evaporation and keyhole mode melting. Lower laser power (30 W), on the other hand, provides melt pools with a low depth-to-width ratio, characteristic of conduction mode melting [40,41,46]. The final structure of the material is largely determined by the thermal history of the HAZ in both scenarios. Typically, during the LPBF process the conduction mode parameters are used as chance of powder ejection from powder bed is reduced, keyhole-induced porosity cannot be formed and residual stresses are mitigated.

The primary crystallization mechanism during the laser processing of MGs, is rather dwell time above glass transition temperature T_g in HAZ than the cooling rate in MP. Furthermore, both heating and cooling rates within the HAZ are nonuniform, affecting the final microstructure of the material [47,48]. These conditions make crystallization control in HAZ the crucial part, thus making it the main focus of LPBF parameters optimization.

Tracks produced within conduction mode are shallow and feature broad crystalline HAZ (C-HAZ) at the MP boundaries. Reducing the laser speed widens the crystalline HAZ, which is typical in MGs LPBF or welding parameter studies [14,41]. On the contrary, tracks melted with keyhole mode have a higher aspect ratio and exhibit less visible C-HAZ, which broadens with a decrease of laser scanning speed. Regardless of the parameters, the HAZ at the bottom of the melt pool retains amorphous (A-HAZ).

The conduction mode is generally preferred during LPBF to minimize spatter generation and achieve high material density. In comparison, the keyhole mode is preferred, considering higher heating rates, to minimize heat input and control crystallization. In a typical LPBF process, the

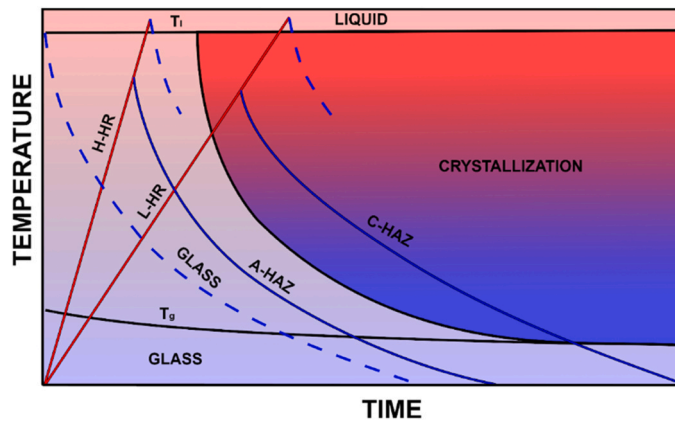


Fig. 4. Crystallization diagram for material heated with the low heating rate (L-HR) and high heating rate (H-HR), resulting in amorphous and crystalline heat-affected zones (HAZ).

microstructure, including phase content and porosity, is a byproduct of laser parameters. To separate densification and glass forming issues, the powder was treated with moderate conduction mode parameters to create bulk material with minimal spatter production [46,49–51], followed by remelting in keyhole mode to establish the final microstructure.

The subsequent experiments were performed with remelting parameters of 3000 mm/s and 120 W to obtain an amorphous structure, or 2000 mm/s and 120 W for a crystalline structure. These parameters were selected based on the HAZ structure observed in the single-track experiments. Same parameters of 2000 mm/s scanning speed and 120 W laser power were chosen to validate the LPBF simulation model. The predicted melt pool depth and width were 195 μm and 94 μm , respectively, as shown in Fig. 3. The coefficients in the Goldak model applied were: $a_f = 1.5e - 5$, $a_r = 1.5e - 5$, $b = 5e - 5$, and $c = 3.1e - 4$. Compared to the experimental data (width: 198 μm and depth: 96 μm), less than 3% of error was achieved for the melt pool size.

Depending on the chosen laser speed, both high heating (H-HR) and low heating rates (L-HR) can be utilized to achieve either an amorphous HAZ (A-HAZ) or a crystalline HAZ (C-HAZ) as shown (Fig. 4). If the material is molten regardless of the heating rate, it is likely to be amorphous due to liquid vitrification (GLASS). In this case, the heating rate (H-HR or L-HR) will determine whether crystalline or amorphous HAZ will form.

3.3. Scanning strategy – global crystallization

Both common laboratory experience and extensive simulation efforts show that the LPBFed material thermal history strongly depends on geometry and scanning strategy [52]. Typically, crystalline materials such as AISI 316 or Ti6Al4V show negligible microstructural change upon overheating the material at the specimen's edge. However, in the context of MGs, surpassing specific heat loads leads to catastrophic devitrification. While there have been reports of heat accumulation and structural relaxation in laser-processed MGs, no MG-specific solution has been suggested [53,54]. Most literature related to AM-MGs focus on volume energy density (VED) as the key determinant of materials porosity and crystalline phase content, with scant attention paid to scanning strategies [55–61].

Past research predominantly concentrates on simple remelting or chessboard strategies, the same as applied to crystalline materials. Lindwall et al. [62] used basic scanning strategies showing that decreasing the hatch distance led to inevitable crystallization, and the processing window depends on the hatch length. These limitations pose challenges for complex geometries, thus necessitating the development of an innovative scanning strategy.

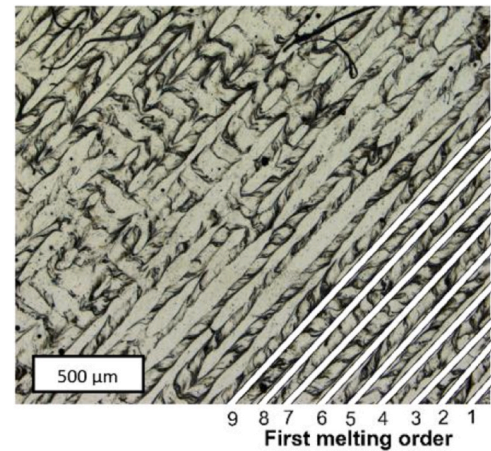


Fig. 5. Optical micrograph of the microstructure of the first melting within conduction mode regime.

Patented procedure [18] was used to melt the powder, effectively avoiding material overheating and maintaining high amorphous phase content. Initial melting was achieved in conduction mode following preliminary optimization with parameters of 30 W, 120 μm hatch distance, and 250 mm/s laser speed at 25 μm slice distance, providing good process stability without extensive splat generation. Each line was adjacent to the subsequential one, with a 90-degree shift in scanning direction for each layer. The microstructure of the material after the initial melting stage and the sequence of scanning lines are shown in Fig. 5. The moderate laser power and speed led to an interpenetrating network of crystalline and glassy phases, with visible Lack-of-Fusion (LoF) type porosity and total porosity under 0.5%, deemed dense for as-printed conditions. It can be observed that the HAZ is severely crystallized despite predominant glass content. The XY pattern is visible due to the partial overlap of subsequent layers.

Post this initial melting, each layer was remelted without powder recoating at 120 W laser power, 30 μm hatch distance, and 3000 mm/s scanning speed. Fig. 6 shows two specimens melted under the same conditions, varying only the sequence of adjacent remelting lines to investigate the influence of heat accumulation on the material microstructure. Specimen A had shuffled line order to increase the time gap between melting adjacent lines, whereas specimen B had lines melted in subsequential order.

The microstructure of the specimens produced with double melting in keyhole mode with shuffled (Fig. 6a) or ordered (Fig. 6b) lines varied, with a visible crystallization in the dark etched field. As the laser navigates towards the specimen's edge, heat accumulation on the top layer leads to global devitrification and catastrophic cracking. The brief interval between subsequently scanned lines ($\sim 300 \mu\text{s}$ for 3000 mm/s scanning speed and 1 mm scanning vector length) results in melt pool interaction and disruption in material cooling. Both specimens show several characteristic keyhole mode spherical pores. Microstructural results align with simulation results which could be an invaluable optimization tool for complex geometries, where heat transfer is a nontrivial issue.

Additionally, from the BMGs welding experiments in LPBF, a significant laser tracks overlap area leads to overheating and crystallization is observed, especially at the edge of the processed material, where heat transfer deteriorates. By changing the melting lines' order to shuffled, global overheating has been significantly decreased. Fig. 7 illustrates a crystallization diagram for shuffled and ordered lines scanning strategy.

The interaction between subsequential lines impedes the material's cooling and prolongs the time when the temperature is in the crystallization-prone range [63,64]. Shuffling the melting lines permits treating each line as an independent thermal cycle, eliminating overlapping and significantly extending the cooling time. Additionally, due

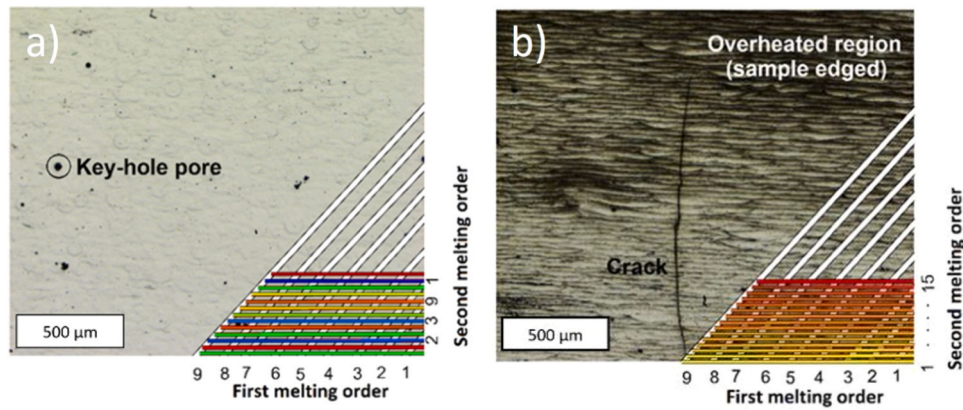


Fig. 6. Microstructure of remelted materials produced with line shuffle strategy (a) and subsequent strategy (b).

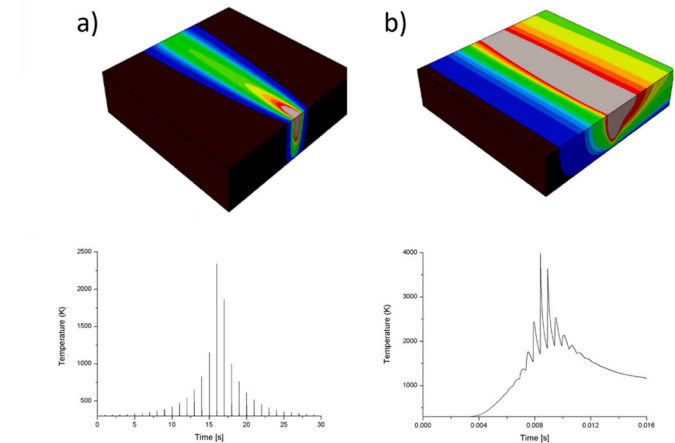
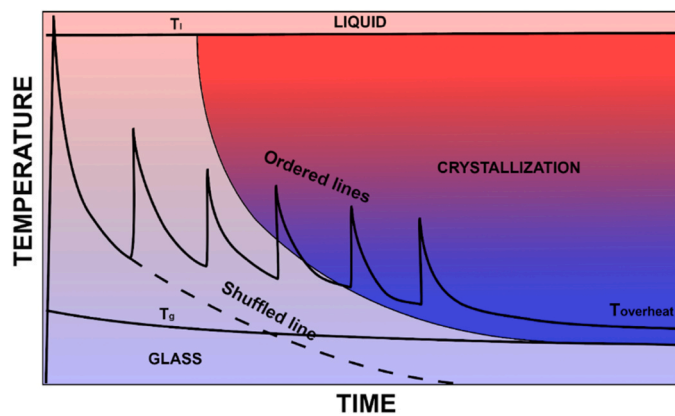


Fig. 7. Schematic illustration of scanning line influence and simulation of heat field of a) line shuffle strategy and b) subsequent strategy.

to the double-stage melting procedure, powder recoating was improved as the surface of the material after first melting was even without deformation at the boundary, which is a usual defect during MGs processing [65]. The flattened material surface achieved during moderate parameters premelting broadens the processing window preventing balling effect and other meltpool-related process instabilities. Overall, the shuffling scanning strategy separates each line into independent thermal events over the whole time of material processing (Fig. 7a), avoiding heat accumulation typical for standard scanning strategies (Fig. 7b). The premelting stage densifies the powder bed with minimum

Table 2
Remelting parameters.

Sample	Laser speed [mm/s]	Power [W]	Hatch distance [μm]	Hatch rotation [degree]	Energy density [J/mm ³]
Reference	250	30	120	90	40
"2000 mm/s 100 μm"	2000	120	100	0	32
"3000 mm/s 100 μm"	3000	120	100	0	19.2
"3000 mm/s 70 μm"	3000	120	70	0	27.4
"3000 mm/s 30 μm"	3000	120	30	0	64

VOD, avoiding issues with powder-to-laser interaction at the second stage.

3.4. Laser melting process optimization

Control of both local (HAZ) and global (heat accumulation) crystallization has been achieved with keyhole remelting laser parameters and shuffled lines order. Having the freedom of phase distribution allowed to prepare a series of composites with adjustable glass phase content by tuning the hatch distance of second laser scanning and laser scanning velocity. The further step involved fine-tuning remelting parameters in forming a laminate-like amorphous-crystalline structure. The reference specimen was melted only once, whereas other specimens were remelted with the parameters listed and the line shuffle strategy, as shown in Table 2, and resultant microstructures are shown in Fig. 9. No hatch lines rotation was involved in the remelting step to keep an anisotropic microstructure.

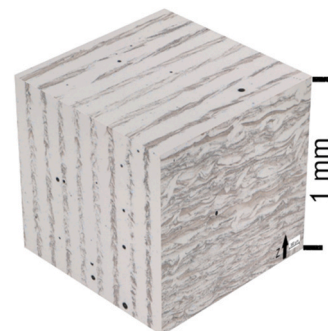


Fig. 8. Microstructure of "2000 mm/s 100 μm" composite.

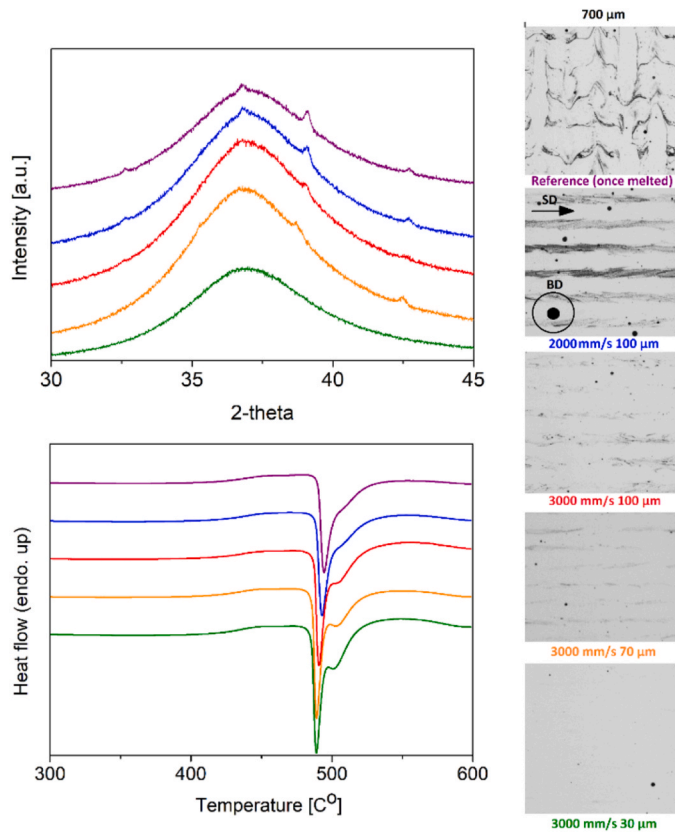


Fig. 9. Preliminary optimization of remelting parameters. BD – build direction, SD – scanning direction. XRD and DSC patterns (left) represent respected microstructures (right).

Both the reference and remelted specimens exhibit porosity lower than 0.5%, with spherical pores located mainly within the melt pools and irregular ones located at MP boundaries. The reference specimen shows the interpenetrating amorphous and crystalline phases, whereas remelted materials have a unidirectional laminate-like structure. Contrary to the previous reports on AMZ4 processing [47,61], the reduction of a hatch distance (H) during the remelting stage providing higher glassy phase content as long as the high laser scanning speed of 3000 mm/s, was kept. The effect is attributed to a more uniform heat distribution achieved with the strategy used. Compared to other parameters studied by Pauly et al. [66] performed with a similar Zr-based alloy, a broader processing window was achieved for amorphous structure up to 64 J/mm² (over 4 times larger than previously reported), keeping near-theoretical density.

The reduction of the laser speed to 2000 mm/s increased the crystallization in HAZ along Scanning Direction (SD), keeping complaint results to single-track experiments as per. Fig. 3 provided a layered, high-oriented microstructure. Fig. 8 depicts the representative microstructure of the "2000 mm/s 100 μm" composite showing the actual anisotropic microstructure. Spherical pores are located in the center of the amorphous zone of the laminate indicating their keyhole-induced nature.

Both X-ray diffraction and differential scanning calorimetry confirm the microstructural observations shown in Fig. 9. The crystalline phase content is barely visible in all the specimens, whereas the specimen "3000 mm/s 30 μm" is X-ray amorphous. The reduction of the hatching distance seems to decrease the crystalline phase content. Cause of this is probably the heat accumulation and increase of the sample temperature. This could be verified by using the thermal camera during the manufacturing process. The diffraction peaks forming on the amorphous peak can be from Zr₂Cu [01–074–7476] or three-component cubic phase

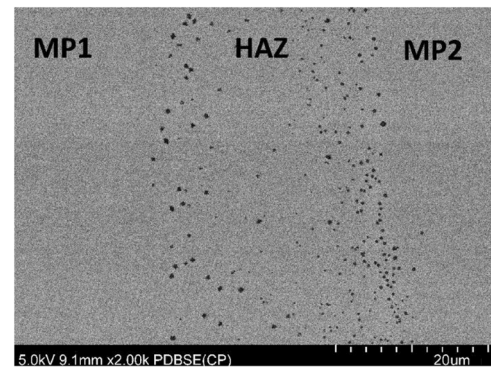


Fig. 10. SEM image of HAZ of 2000 mm/s 100 μm specimen. Black dots are crystallites from which TEM was conducted.

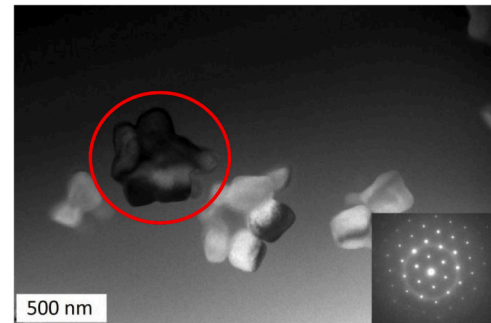


Fig. 11. TEM image of 2000 mm/s 100 μm specimen HAZ with selected area electron diffraction pattern in the orientation $z = [111]$.

ZrCuAl [04–018–3885] or [04–020–8421] [32].

Lindwall et al. [48] showed that the crystallite size is dependent on their position within HAZ - larger (due to higher growth rate) crystallized near the fusion zone and numerous smaller ones (due to higher nucleation rate) at the edge of the HAZ. Due to the MP overlapping, this effect is hard to observe in the presented specimens. Nevertheless, considering the order of the melt pools, it can still be distinguished in Fig. 10.

TEM investigations of "2000 mm/s 100 μm" HAZ evidenced dendritic submicron crystallites within the amorphous matrix. As shown in Fig. 11, dendrites of less than 1 μm were observed during the microstructure observations. Electron diffraction of these crystallites revealed a cubic Cu₂OZr₄ phase structure in the [111] direction, whereas no α-Zr (O) nanocrystals were observed, similar to previous reports by Marattukalam et al. [67]. A clear halo from the residual amorphous phase is also visible at the diffraction.

The small volume fraction of crystallites shown via SEM explains the weak XRD signal, whereas light microscopy of the etched specimen tends to overestimate crystalline phase content. Usually, low hatch distance leads to substantial crystallization within HAZ, but the effect is reversed in the specimens processed with a "shuffled line" strategy. Considering the width of the premelting HAZ (Table 1), such a glassy phase increase cannot be explained with a purely geometrical relation. The remelting hatch distance should cover at least the HAZ created

Table 3
Oxygen and nitrogen concentration in powder and composites.

Sample	O ₂ (ppm wt.)	N ₂ (ppm wt.)
Powder	1762 ± 40	47 ± 5
Composite	1990 ± 82	78 ± 4
Amorphous	1900 ± 45	72 ± 2
Once melted	1855 ± 70	60 ± 2

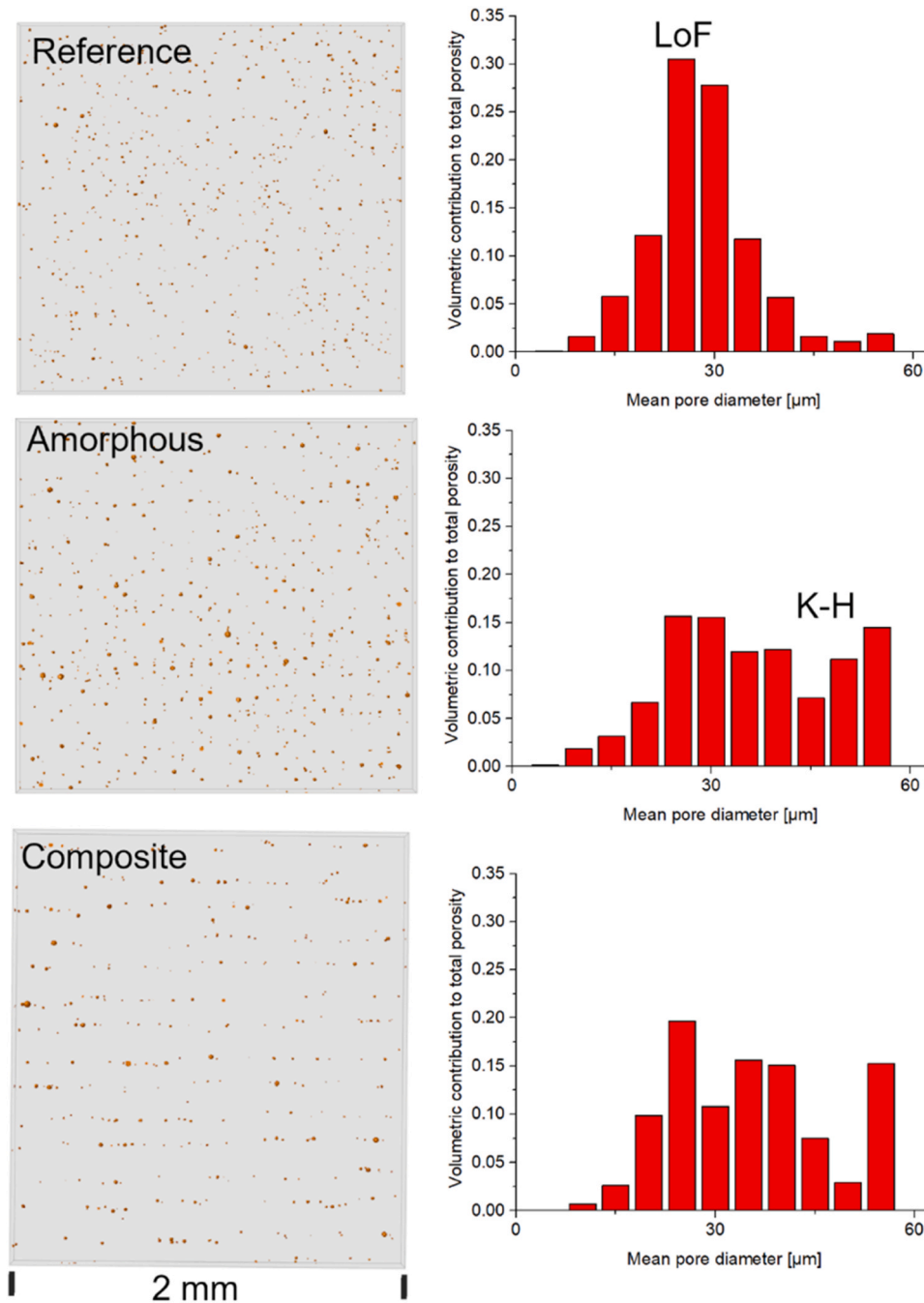


Fig. 12. CT images and pore distribution of the reference, composite, and amorphous specimens. LoF Lack-of-Fusion pores, K-H keyhole pores.

during premelting to provide a uniform amorphous structure. However, the material has to be remelted multiple times as depth of the meltpool depth (140 μm) exceeds the powder layer thickness (25 μm) and the meltpool width (85 μm) exceeds the hatch distance (30 μm). It suggests that the melting of crystallites in HAZ created during premelting could be a complex phenomenon dependent on energy input or chemical homogenization. It is worth noting that additional melting stages increased the impurities level in the final material, as shown in Table 3, yet it is a minor effect compared to changes in the thermal history of the material.

Further tests were realized with "3000 mm/s, 30 μm" parameters chosen to be the AMORPHOUS benchmark and "2000 mm/s, 100 μm" chosen to be the COMPOSITE benchmark.

3.5. Porosity reduction via high-pressure heat treatment

There are numerous sources of porosity in LPBF, such as powder moisture, evaporation of volatile elements, keyhole instability, and poor wettability of subsequential layers. All those sources result in porosity, which has been an inevitable part of MGs processing until now. To reduce the porosity of the amorphous and composite materials, several hot isostatic pressing cycles were performed with a state-of-the-art high-pressure Hot Isostatic Press (HIP – IHPP PAS "UNIPRESS", Poland). Process was conducted using following parameters: heating rate 40 K/min, holding temperature of 420 °C, holding time of 360 s, and pressure of 5.0 Ar 1 GPa. Parameter selection was based on the previous preliminary studies to simulate the SPS conditions in which it was possible to obtain amorphous AMZ4 gears[32]. The gas pressure was set to the

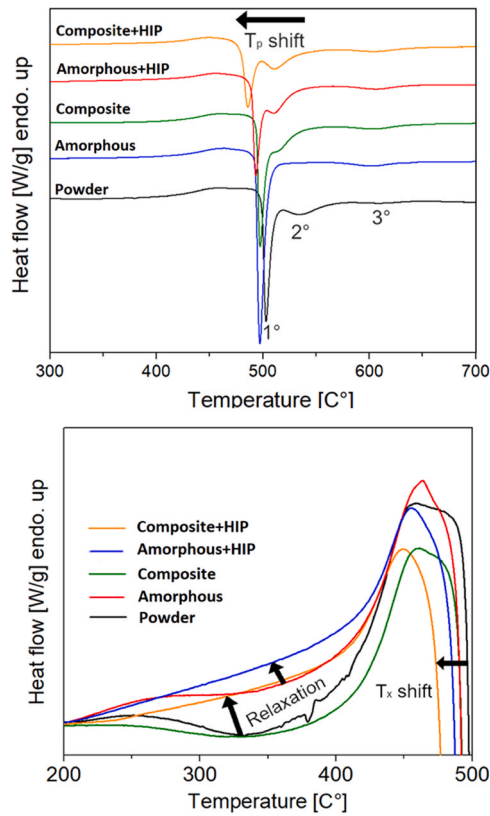


Fig. 13. DSC traces for hipped specimens T_p – event peak temperature, T_x – crystallization temperature onset.

maximum value of the used equipment. As printed and HIPed specimens were investigated by CT with pores distribution plotted against pore size, XRD and DSC measurements. Due to possible crystallization and variations in theoretical densities for MGs at different relaxation stages [68] CT was used as a primary density measurement method instead of Archimedes' method. The CT reconstruction of specimens is shown in Fig. 12. The REFERENCE (once melted), AMORPHOUS, and COMPOSITE specimens have porosity near 0.5% of the total specimen volume. The reference specimen (melted once) had normal pore size distribution with a maximum of 25 μm mean pore diameter with no oriented distribution. Specimens melted twice with higher laser power have highly oriented porosity with larger pores of 60 μm mean pore size caused by the keyhole instability. The composite specimen has layered pore distribution accordingly to the hatch distance (100 μm). The amorphous specimen has a characteristic pattern of equidistant pores, which could be formed by repetitive single-point laser exposure. It is worth noting that the laser scanning lines are not continuous, yet it rely on decreet values of distance between the adjacent point and exposure time within the point. Melt pool instabilities connected to a discreet laser scanning model could be investigated further by the in situ methods [69].

The HIP procedure reduced porosity beyond the resolution of the used CT in an AMORPHOUS specimen, whereas in a similar COMPOSITE specimen, only three (3) pores were found in the CT scan. As the specimens were processed at ultrahigh pressure, the densification is attributed to an accelerated material flow and diffusion in the supercooled region. The AMORPHOUS specimen remained amorphous during the HIP procedure while the COMPOSITE specimen underwent further crystallization, as shown in Fig. 13.

In order to identify the phase composition and crystal structure of the phases occurring in the printed specimens, synchrotron X-ray diffraction was performed. The diffraction patterns were taken for the additive manufactured specimens and treated with the HIP process. In the case of both initially amorphous specimens, the diffraction patterns show only

Table 4

Characteristic temperatures in DSC analysis.

Sample	T_x (°C)	T_{p1} (°C)	T_{p2} (°C)	T_{p3} (°C)	$-\Delta H$ (°C)
Composite+HIP	478.9	484.7	509.9	602.6	57.6
Composite	493.2	496.3	509.9	603.2	65.2
Amorphous+HIP	488.1	492.21	509.3	601.3	63.4
Amorphous	491.9	496.0	X	600.6	64.6
Powder	497.5	501.8	534.51	604.76	64.2

two halos at 2 theta = 3.35° (1.218 Å) and 5.69° (0.718 Å). However, in the case of the COMPOSITE specimens, some trace crystalline lines were observed. The crystallinity of the COMPOSITE sample was estimated to be 2.7%. The HIP process not only reduces the porosity of the COMPOSITE sample but also increases it to about 7.1% primarily showing the $\text{Cu}_2\text{Zr}_4\text{O}$ phase.

As shown in Fig. 13, both AMORPHOUS and COMPOSITE specimens underwent relaxation during HIP. It is intuitional since relaxation is the free volume reduction leading to atomic densification and high pressure should facilitate this phenomenon. Characteristic temperatures such as T_g (glass transition), T_x (crystallization onset), and T_p (event peak) are lower after HIP for both materials (Table 4). The T_x reduction and separation of the first and the second thermal effects are attributed to higher crystalline phase content. The crystallinity of the COMPOSITE specimen increased during HIP due to the presence of LPBF-induced nuclei. Crystallization enthalpies (first and the second effect combined) are similar for all the investigated parameter sets except "COMPOSITE+HIP" which has a crystallization enthalpy about 10% lower than other specimens. Partial overlapping of the two thermal events creates certain difficulties in the reliable measurement of crystallization enthalpy, yet visibly lower T_x and T_x of HIP-processed specimens indicate less stable structures.

In Fig. 14, two effects are visible: pore reduction and a slight difference in Nomarski contrast, which is extremely sensitive to height differences in the range of a tenth of a nanometer. Amorphous melt pools reveal complex structures suggesting either a difference in medium-range order (MRO) or dealloying during remelting. After the HIP step, no such effect is visible, suggesting that the MRO hypothesis could be investigated further even within a single melt pool with respect to recent layer-based studies by Best et al. [27].

3.6. Mechanical properties of amorphous and composite specimens

In the next step, the bending strength of the as-printed and HIP materials was investigated. The specimens were prepared with the abovementioned laser melting and heat treatment parameters using the very same powder batch. The 4-point bending test was similar to those reported previously by Bordeenithikasem et al. for PBF AMZ4 [15], with force applied perpendicular to Build Direction, specimens dimensions of 2x3x25 mm, and bent either perpendicular (90°) or parallel (0°) to the rescanning vectors, as shown in Fig. 15.

Bending strength results are summarized in Table 5. No macroscopic plastic deformation was observed in any case. The as-printed, AMORPHOUS specimens had similar strength in both load directions (90° - 1900 $\pm$ 90 MPa; 0° - 1550 $\pm$ 230 MPa). Also in both cases, HIP decreased their strength (90° - 1500 $\pm$ 270 MPa; 0° - 1090 $\pm$ 190 MPa). The highest strength was achieved for the as-printed, AMORPHOUS material, bent perpendicular to the rescanning vector, whereas pore-free material after HIP, had lower strength, disproving porosity as the main factor that determines the strength of PBF MGs [15,66]. Lower strength of AMORPHOUS samples can be attributed to relaxation of the amorphous phase during the HIP. For the as-printed COMPOSITE, the strength was strongly dependent on load direction (90° - 1640 $\pm$ 40 MPa; 0° - 560 $\pm$ 50 MPa), and the difference was even more considerable after HIP (90° - 1800 $\pm$ 30 MPa; 0° - 150 $\pm$ 30 MPa). Ultimate strength was deteriorated by order of magnitude only by changing the loading

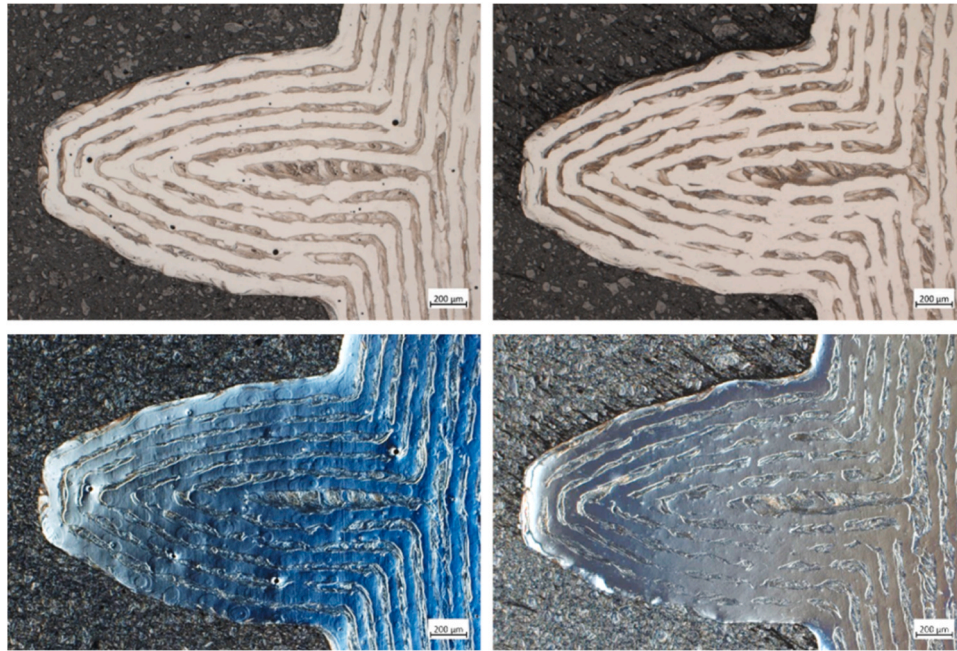


Fig. 14. Composite specimens before (left) and after HIP (right) - bright field (top) and Nomarski contrast (bottom).

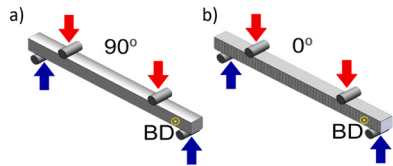


Fig. 15. Specimen orientation during bending. BD-build Direction, 90°-perpendicular bending, 0° - parallel bending.

direction of heat-treated COMPOSITE. In every case, the COMPOSITE specimens had much lower strength results scattering than the AMORPHOUS specimens. AMORPHOUS specimens exhibited higher bending strength than the COMPOSITE ones, which is not surprising due to the fragility of crystalline phases in the presented system [70]. This fragility also significantly reduced the bending strength of the COMPOSITE after HIP when measured in a parallel direction (0°), where crystalline layers fractured very easily from the tension side of the sample. However, the bending strength of the COMPOSITE sample after HIP measured in the perpendicular direction (90°) was not only higher than before densification but also higher than the AMORPHOUS benchmark after HIP.

Crystalline phase fragility did not play a vital role in the parallel direction (0°) as there are no apparent weak spots at the tension side of the specimen. Additionally, the crystalline layers seemed to be shear band barriers hindering their catastrophic propagation, typical in room temperature strength tests [71]. This shear band blocking effect could have been strengthened by the porosity reduction during HIP and an increase of crystalline phase content, improving mechanical contact between amorphous and crystalline layers and increasing the number of shear band barriers.

Detailed fractography was performed for the 90° and

0° COMPOSITE+HIP specimens (Fig. 16). Etched crystallites are visible at the fractography revealing layered microstructure. The fracture of the 0° specimen is smooth and featureless, showcasing a brittle crack through the crystalline phase. For 90° orientation a secondary crack is visible on the first low magnification photo, showing a crack tip deviation at boundaries between the crystalline and amorphous layers. The 90° fracture shows a characteristic vein-like pattern-a relic of microscopic MG plastic deformation as well as retained shear bands, which delayed the catastrophic propagation of cracks.

To assess the influence of the COMPOSITE orientation on wear resistance, new specimens in the shape of 25 mm diameter disks were manufactured for the pin-on-disc tests. Their glassy/crystalline patterns are shown in Fig. 17. Significant differences in the coefficient of friction were observed between the specimen in which the crystal paths were perpendicular to the direction of motion of the zirconium oxide beads (aka BICYCLE pattern), and the specimen in which the paths were parallel (aka ONION pattern). In the case of the BICYCLE specimen, fluctuations in the coefficient of friction can be observed, and it remains higher than for the ONION specimen.

As shown in Fig. 18, differences are also evident in debris accumulation. The BICYCLE specimen is characterized by a shallower wipe mark and more material embedded in the surface than the ONION specimen. At higher magnifications, one can see the different wear mechanisms between the specimens. Three features are visible: plastically deformed grooves, embedded particles, and wear products. Zirconium oxide particles are pushed into the specimen surfaces for the ONION pattern, while a mixture of zirconium oxide and substrate material forms deposits in the BICYCLE pattern. It is reasonable to assume that the presence of transverse obstacles in the first specimen promotes the formation of deposits, which limits further wear and results in a variable coefficient of friction reading over time. Compared to other studies involving pin on disk tests [15,70], results presented for the

Table 5
Strength of bent specimens.

State	AMORPHOUS				COMPOSITE			
Load direction	(90°)		(0°)		(90°)		(0°)	
Treatment	None	HIP	None	HIP	None	HIP	None	HIP
Bending strength [MPa]	1900 ± 90	1500 ± 270	1550 ± 230	1090 ± 190	1640 ± 40	1800 ± 30	560 ± 50	150 ± 30

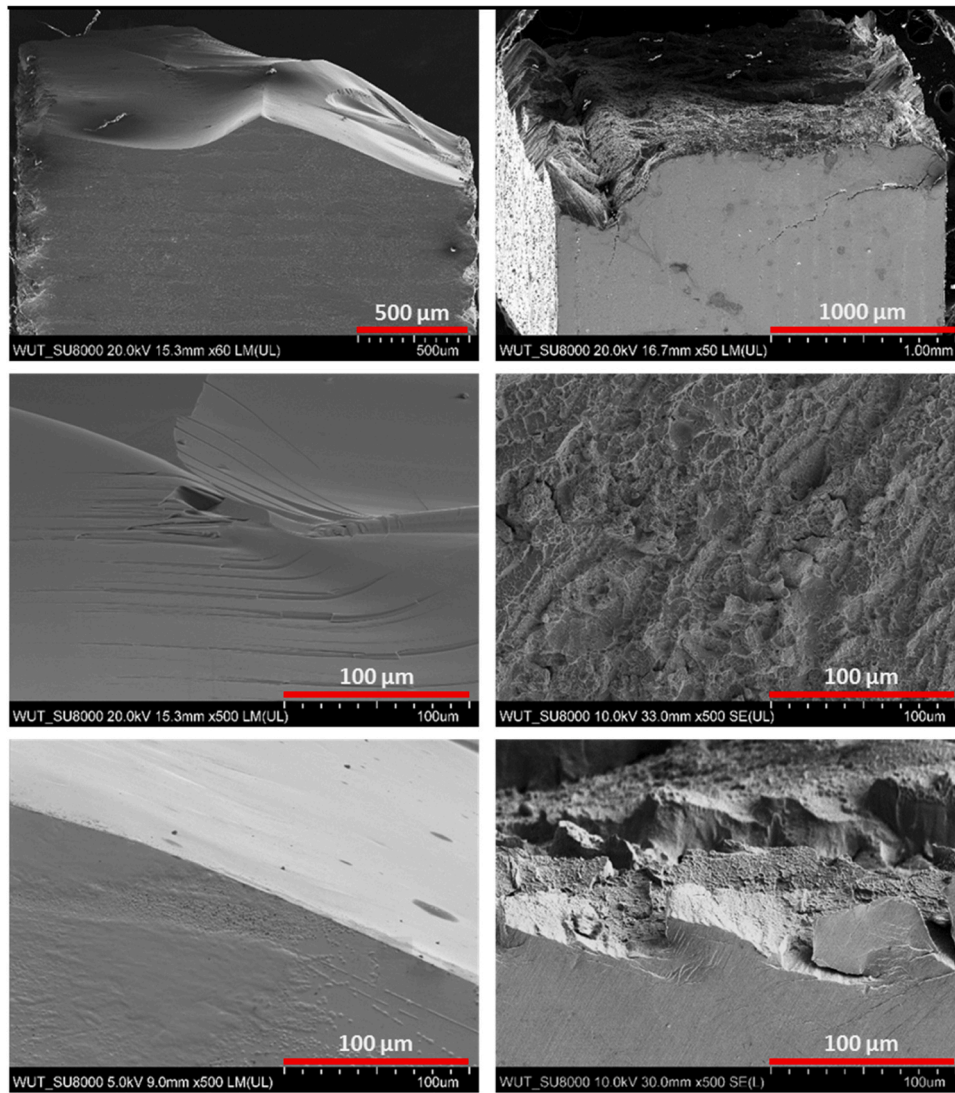


Fig. 16. SEM images at different magnifications of fractography of COMPOSITE after HIP at 0° (left) and 90° (right).

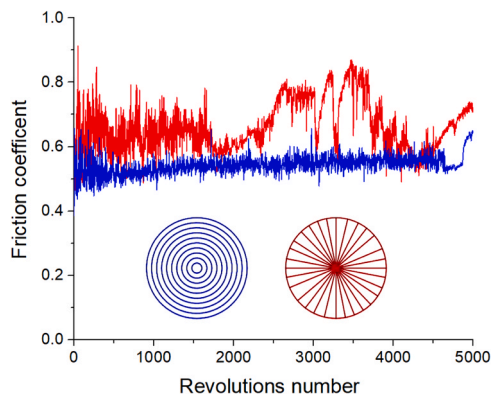


Fig. 17. Friction coefficient for as-printed COMPOSITE.

ONION variant of COMPOSITE material had a surprisingly stable coefficient of friction, whereas the BICYCLE one seemed to accumulate debris.

To confirm this assessment, the nano-hardness mapping was performed for the COMPOSITE specimens. Fig. 19 displays the respective nano-hardness map. The Scatter of nano-hardness value was 4–8 GPa

excluding pores. Hardness distribution corresponds to the microstructural formation with 8 GPa hardness in HAZ and 4 GPa in the melt pool. Such behavior could result from crystallization and glass relaxation as despite being fully amorphous MPs are not uniform in hardness as reported previously by Li et al. [57]. Therefore, the wear resistance can be designed by the spatial distribution of hard-crystalline phases.

4. Future outlook and application

To fully utilize the potential of in situ MGMCs, produced with LPBF and HIP it is essential to design the microstructure in such a way that the bending direction and wear direction of the final component are perpendicular to glassy-crystalline layers. Two of such examples were shown in Fig. 20– high-performance gear with phase distribution designed along with tooth and knife with a designed composite structure. Both parts were created by overlapping multiple thin-walled STL models and laser melted with optimized COMPOSITE parameters at the boundary of the STL model. The gear example was additionally processed with HIP no porosity is visible on the polished surface. The knife example cracked in the middle shows that LPBF-related stresses are still a challenge for the process.

The abovementioned scanning strategy can be quickly transferred to other nonequilibrium systems. AMZ4 is widely available, yet its primary

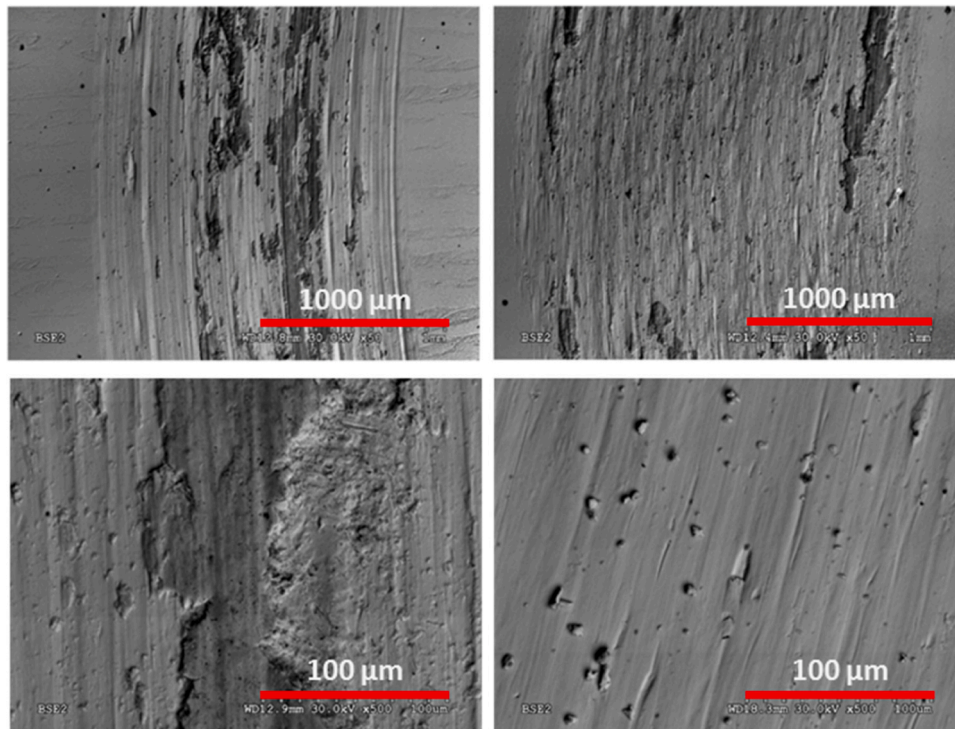


Fig. 18. SEM surface of the COMPOSITE after pin-on-disc wear test BICYCLE (left) and ONION pattern (right).

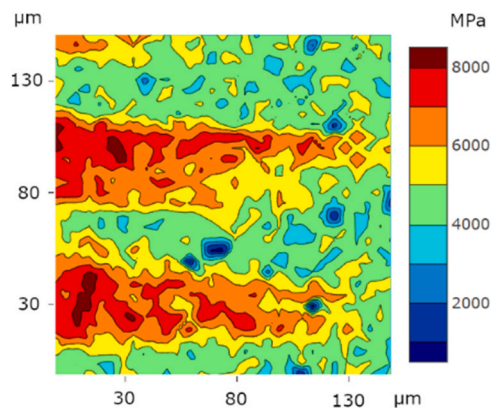


Fig. 19. Nano-hardness mapping of a COMPOSITE specimen.

crystalline phases are brittle, while other alloys could benefit from crystalline phase properties. The equiatomic ZrCu is a particularly interesting material due to the possibility of strengthening via B2 phase transformation. Such a material has been atomized with the ultrasonic procedure [72] and powder below 63 μm has been processed with the same parameters as for REFERENCE, AMORPHOUS, and COMPOSITE material described in the paper (Fig. 21). As shown in the micrograph, a similar structure can be achieved for AMZ4 even though no extensive optimization was involved. The powder was initially crystalline, yet after remelting, an almost entirely amorphous or composite structure can be obtained proving that the described strategy can be easily transferred to other systems (Fig. 22).

5. Conclusions

1. The formation of a crystalline phase in Metallic Glasses (MGs) processed through Laser Powder Bed Fusion (LPBF) is attributable to devitrification occurring within the Heat Affected Zone (HAZ).

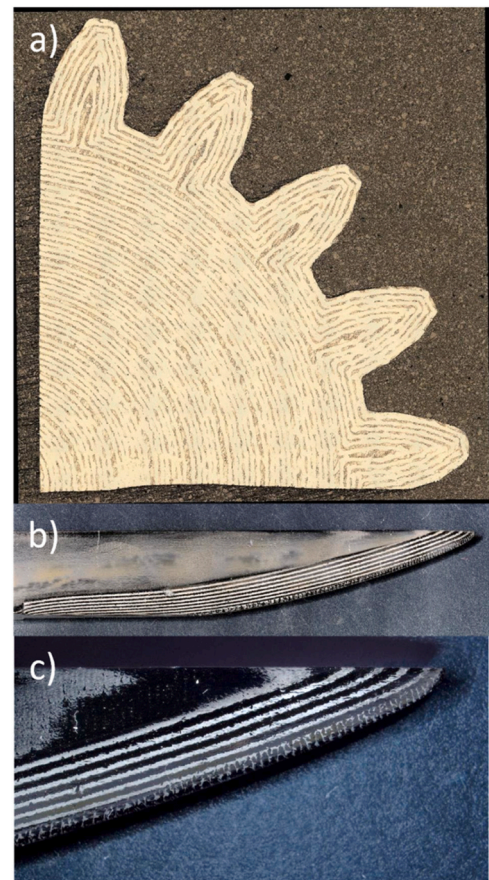


Fig. 20. Application examples – a) composite gear b) knife c) knife closeup.

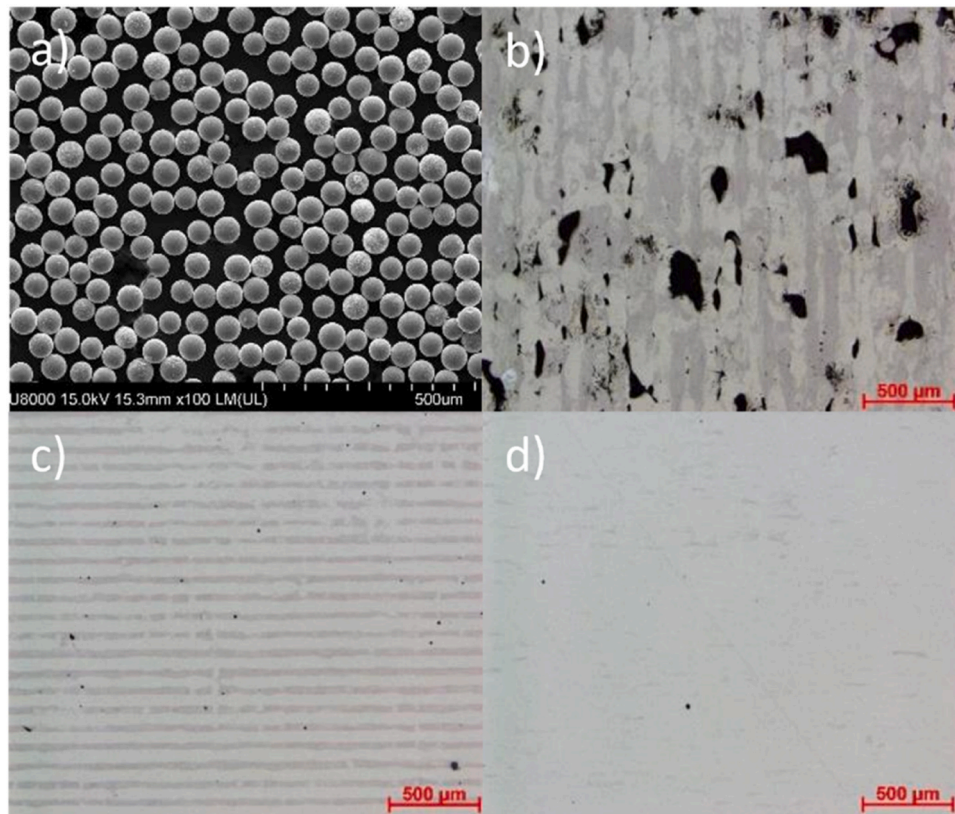


Fig. 21. Pictures of a) ZrCu powder feedstock, b) material melted with REFERENCE parameters, c) material melted with COMPOSITE parameters, d) material melted with AMORPHOUS parameters.

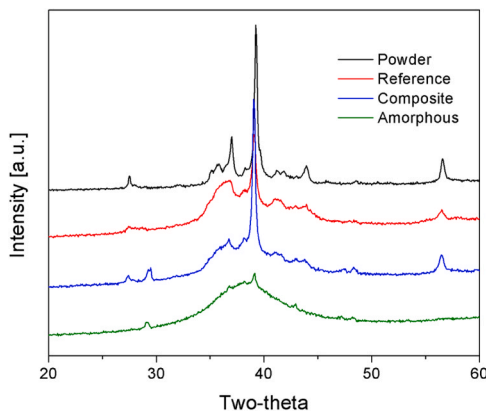


Fig. 22. XRDs of ZrCu powder and processed specimen.

Significant heating and cooling rates result in an amorphous structure, making keyhole mode melting - facilitated by a high scanning speed - an optimal melting mode. Reducing scanning speed leads to crystalline-HAZ creation.

2. Using a double-stage melting approach that incorporates both a pre-melting stage and a line separation strategy allows for the production of nearly dense MGs and MG matrix composites (MGMCs), averting excessive heat build-up within the material. This two-step scanning approach segregates the interaction between powder and laser, as well as spatters generation issues, from the ultimate densification and microstructural design phases.

3. Contrary to the single-melting strategy, double-stage melting necessitates the melting of a previously devitrified HAZ. Therefore, the degree of amorphization can be enhanced by increasing the Volumetric

Energy Density.

4. MG and MGMCs can be fully densified via Hot Isostatic Pressing (HIP). A carefully designed process can prevent MGs from further crystallization, although they undergo structural relaxation. In MGMCs, crystallization continues on LPBF-induced nuclei.

5. LPBF-produced MGMCs may exhibit considerable mechanical anisotropy, with bending strength differences exceeding an order of magnitude being observed when HIP was incorporated. The wear resistance and bending strength of these MGMCs are highly reliant on the phase distribution.

The established guidelines for the LPBF process can be considered universally applicable to both MGs and MGMCs, as demonstrated by the ZrCu example. In conclusion, it's noteworthy that the discussed three-step strategy (powder sintering, remelting for microstructural design, and HIP) effectively extends the well-known Design for Additive Manufacturing (DfAM) principles to encompass both geometry and microstructure design simultaneously.

Funding

The authors acknowledge financial support from the Polish Ministry of Science and Higher Education with "Diamentowy Grant" 0213/DIA/2017/46 and Polish National Centre for Research and Development (grant number TECHMATSTRATEG2/41094/4/NCBR/201).

Authors statement

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

CRediT authorship contribution statement

Blyskun Piotr: Investigation, Data curation. **Yuan Lang:** Writing – original draft, Visualization, Validation, Software. **Świąszkowski Wojciech:** Writing – review & editing, Methodology, Investigation. **Li Xiaopeng:** Writing – original draft, Visualization, Software. **Ciftci Jakub:** Investigation, Writing - review & editing. **Chulist Robert:** Methodology, Data curation. **Ura-Bińczyk Ewa:** Methodology, Formal analysis. **Małachowska Aleksandra:** Visualization, Data curation. **Dobkowska Anna:** Investigation. **Wysocki Bartłomiej:** Validation, Formal analysis, Conceptualization. **Leonowicz Marcin:** Writing – review & editing, Validation. **Kulikowski Krzysztof:** Formal analysis, Data curation. **Wróblewski Rafał:** Methodology, Conceptualization. **Moneta Grzegorz:** Visualization, Data curation. **Choma Tomasz:** Writing – review & editing. **Cetner Tomasz:** Validation, Methodology, Investigation. **Morończyk Bartosz:** Writing – original draft, Investigation, Formal analysis. **Źródowski Cezary:** Visualization, Conceptualization. **Krawczyńska Agnieszka:** Conceptualization. **Źródowski Łukasz:** Writing – original draft, Validation, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Jaroszewicz Jakub:** Validation.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Łukasz Źródowski reports equipment, drugs, or supplies was provided by AMAZEMET. Łukasz Źródowski reports a relationship with AMAZEMET that includes: board membership and equity or stocks. Bartosz Morończyk, Tomasz Choma, and Jakub Ciftci report a relationship with AMAZEMET that includes: employment. Łukasz Źródowski has patent #PL238391 issued to Łukasz Źródowski. Łukasz Źródowski has patent #WO2020230027 pending to Łukasz Źródowski. author previously owned shares of company 3D-Lab sp. z o.o. - LZ author patented a method of BMGs production via L-PBF which was sold to Heraeus GmbH – LZ.

Data availability

Data will be made available on request.

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